

CSCI-4360/5360

Web Technologies

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Lecture on:



Whale writes an app on whale's computer



Whale writes an app on whale's computer



**Other whale download the app
on other whale's computer**

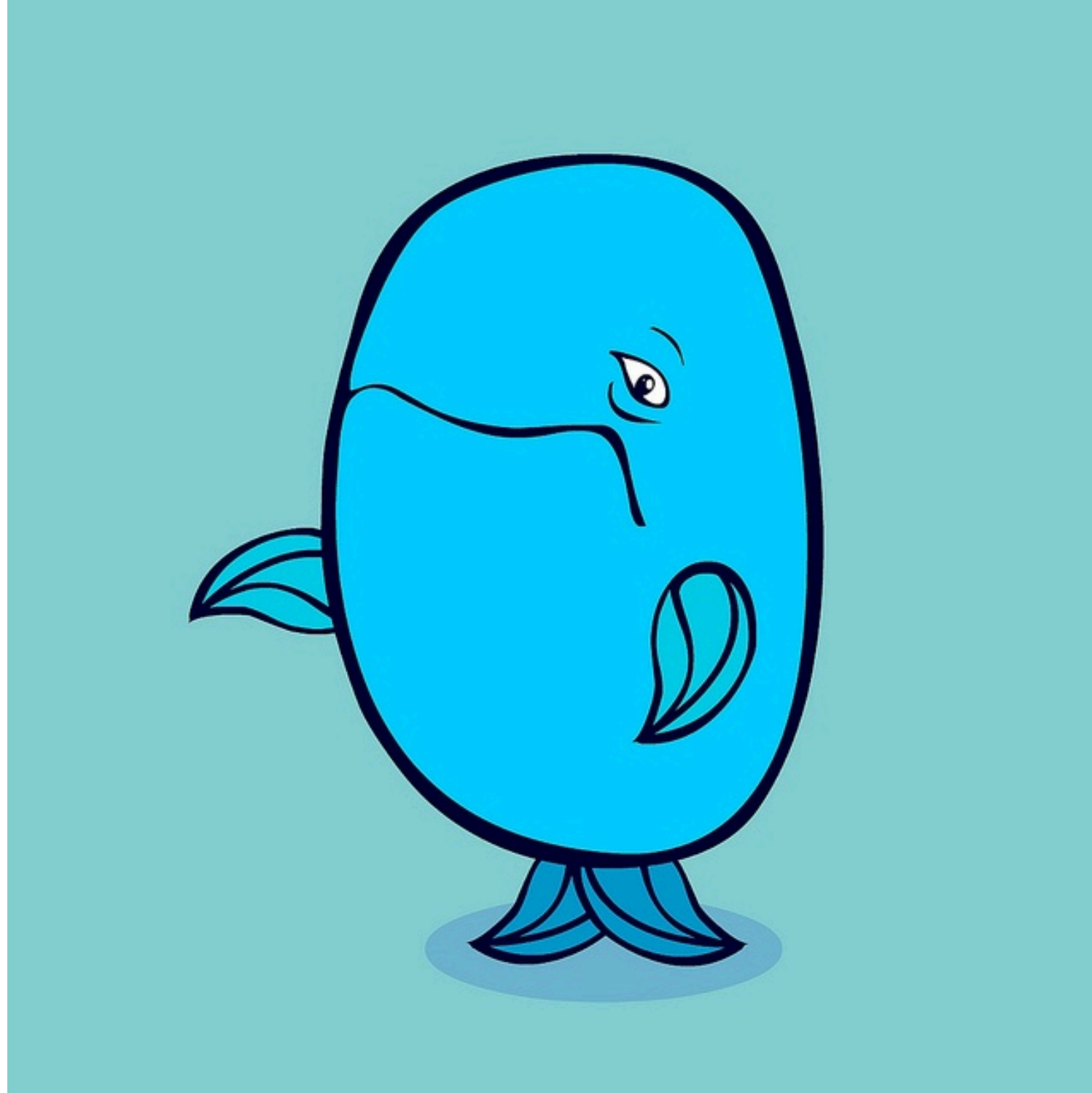


**Other whale download the app
on other whale's computer**

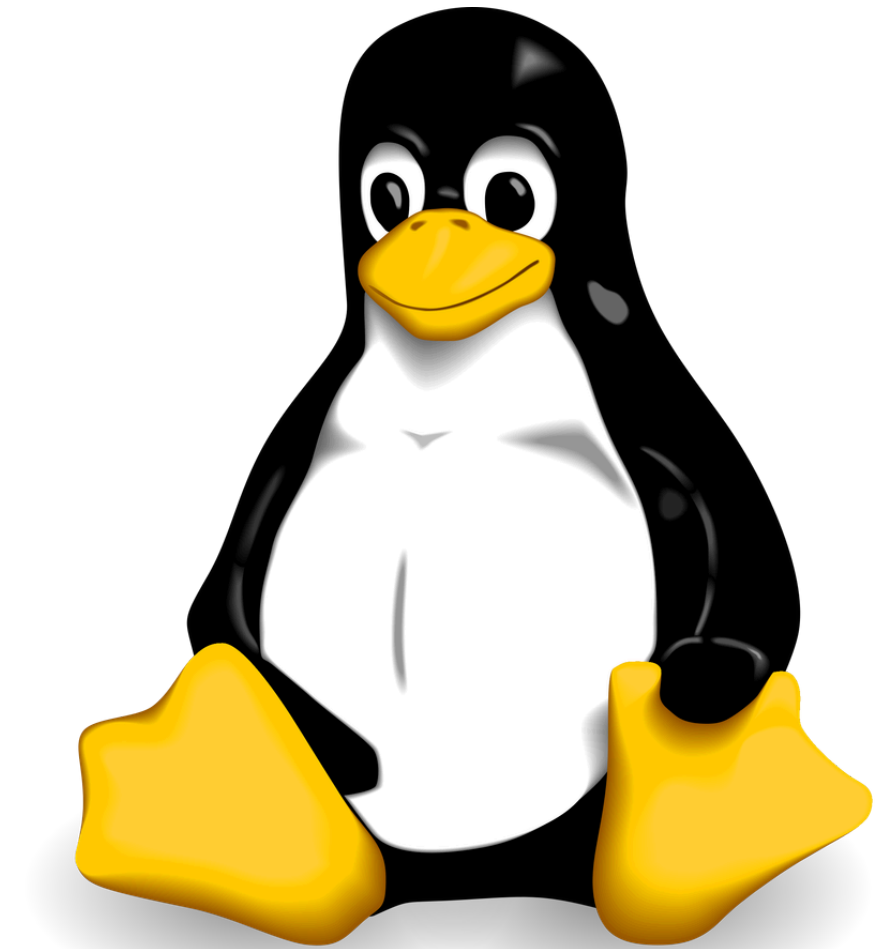
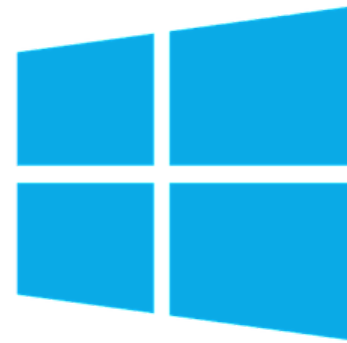


The app breaks

Why?



Why?

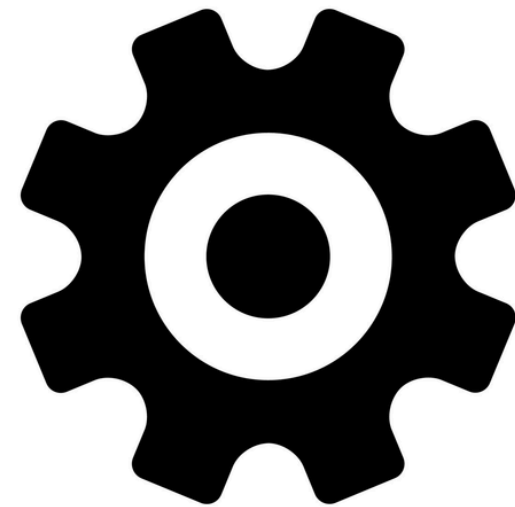


Different enviroment

Smart Whale add a docker file to the app



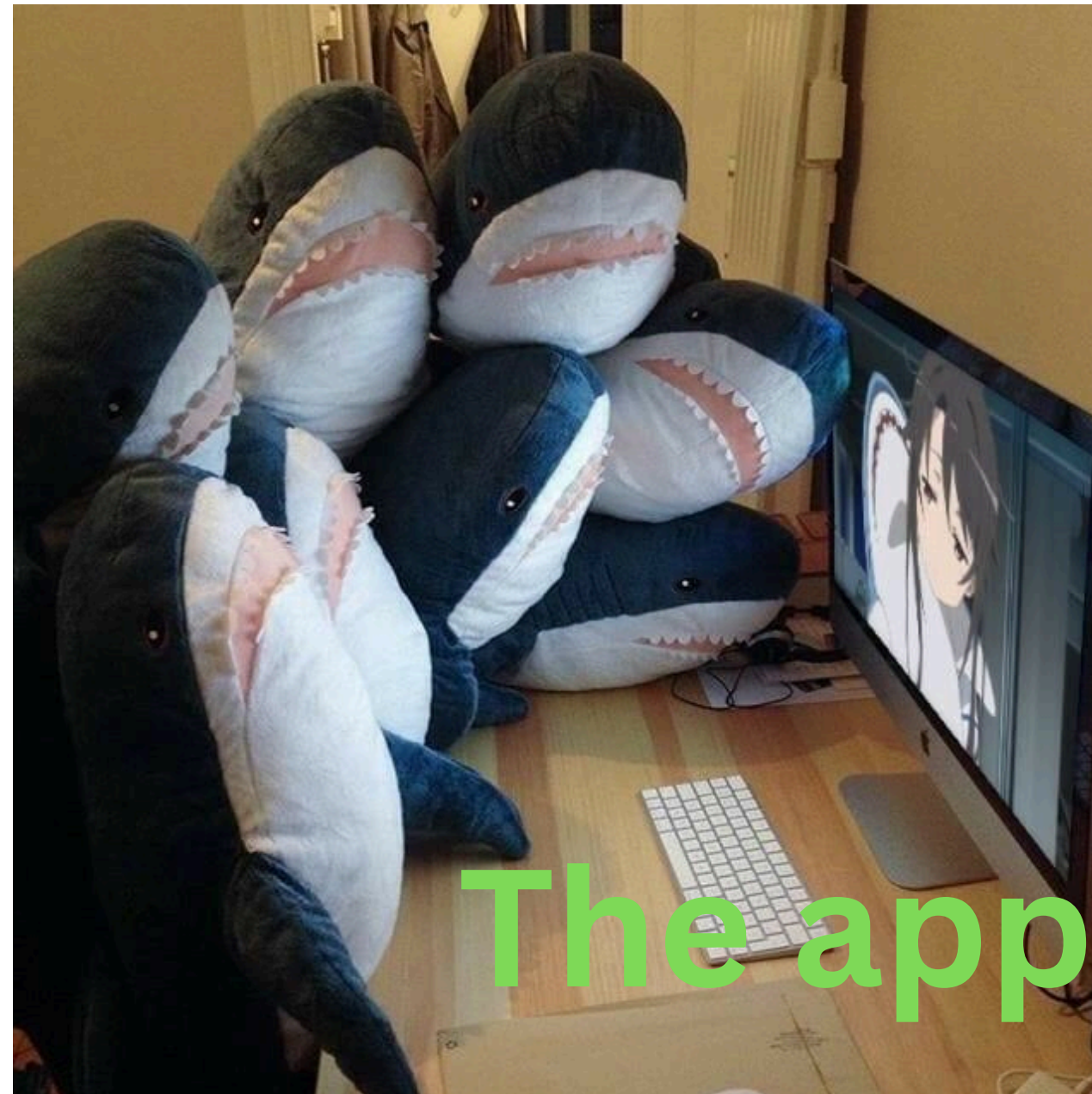
What is in the Docker file?



**Another whale runs the app on
another whale's computer with
the Docker file**



**Another whale runs the app on
another whale's computer with
the Docker file**



The app works

What has changed?

What has changed?

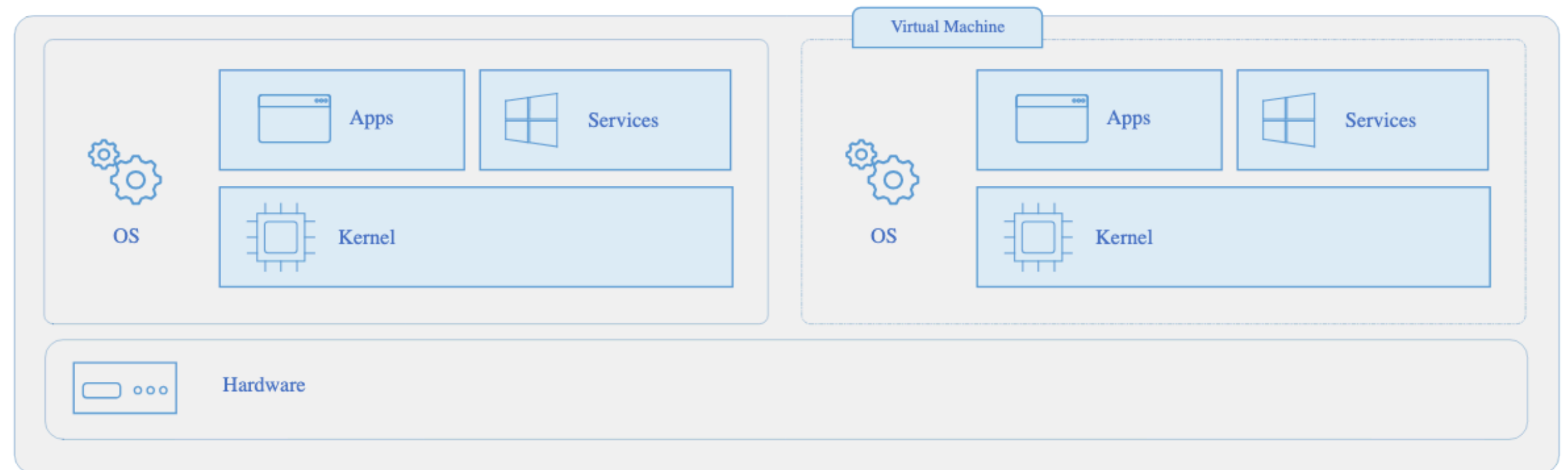
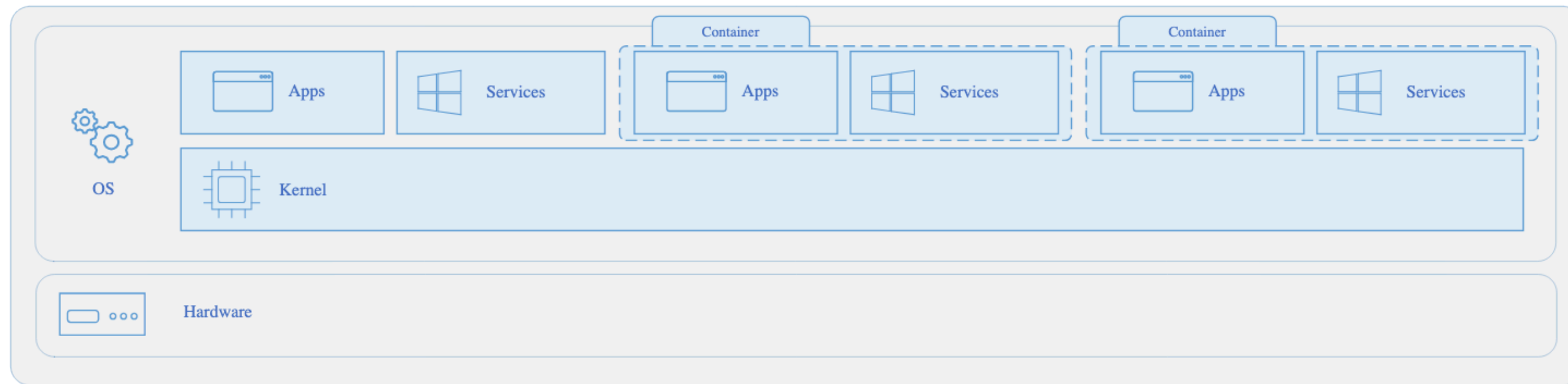
Docker makes the environment
the same anywhere

What is Docker?

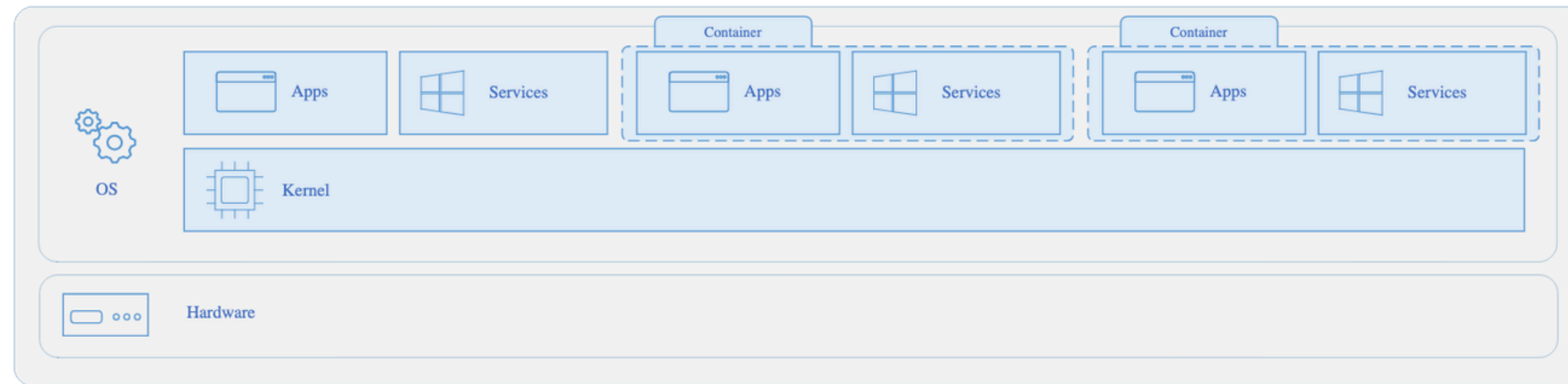
Linux Containers, in general, and technologies like Docker and Kubernetes, are:

a set of Unix libraries designed to *build, run, and “ship”*
isolated applications

VMs vs Containers: What is the difference?

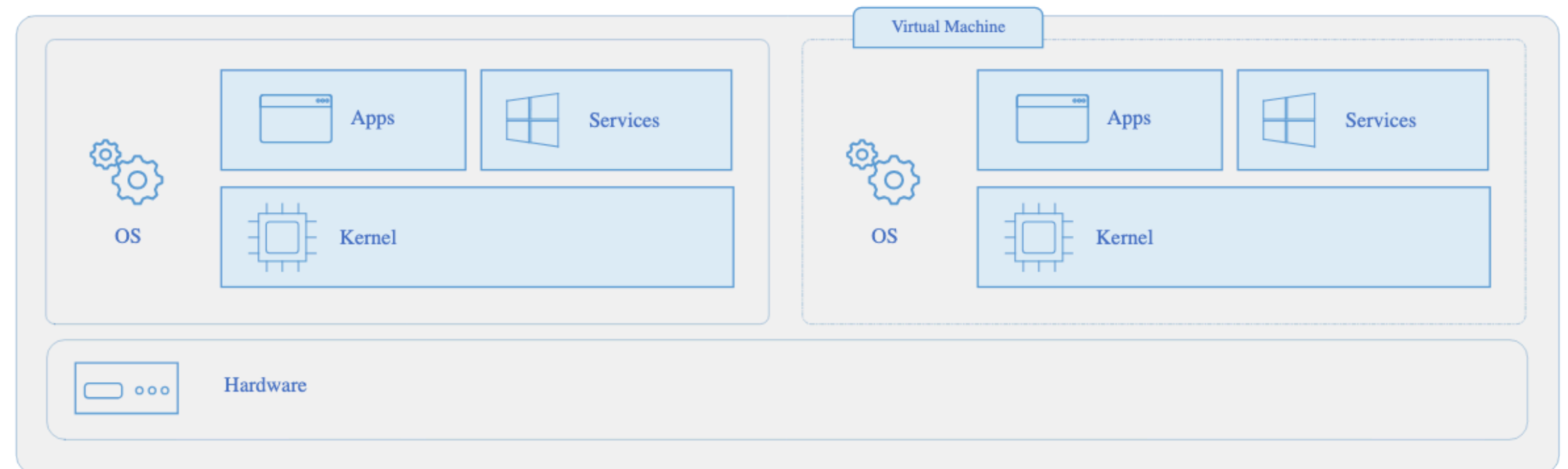


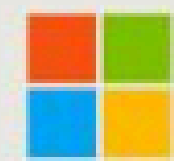
VMs vs Containers: What is the difference?



Container

Virtual Machine





Microsoft Learn

Containers vs. virtual machines

Overview of the key similarities and differences between containers and virtual machines, and when you might want to use each.



MicrosoftLearn /

Image vs Container

Docker Image	Docker Container
It is a blueprint of the Container.	It is an instance of the Image.
Image is a logical entity.	The container is a real-world entity.
Images are created only once.	Containers are created any number of times using an image.
Images are immutable. One cannot attach volumes and networks.	Containers change only if the old image is deleted and a new one is used to build the container. One can attach volumes, networks, etc.
Images do not require computing resources to work.	Containers require computing resources to run as they run with a Docker Virtual Machine.
To make a docker image, you have to write a script in a Dockerfile.	To make a container from an image, you have to run the "docker run <image>" command

Image vs Container

Docker Image	Docker Container
An image must not refer to any state to remove the image.	A container must be in a running state to remove it.
One cannot connect to the images as these images are like snapshots.	In this, one cannot connect them and execute the commands.
Sharing of Docker Images is possible.	Sharing of containers is not possible directly.
It has multiple read-only layers.	It has a single writable layer.
These image templates can exist in isolation.	These containers cannot exist without images.
Docker Images are used to package up applications and pre-configured server environments.	Containers use server information and a file system provided by an image in order to operate.

Let's play with Docker

app.py

```
print("Hello from inside Docker!")
```

Docker file

```
FROM python:3.11  
COPY app.py /app.py  
CMD ["python", "app.py"]
```

Dockerfile

Build the image

```
docker build -t mypythonapp .
```

Run the container

```
docker run mypythonapp
```

Docker compose for running multiple containers together

```
version: "3"
services:
  web:
    image: nginx
    ports:
      - "8080:80"
```

docker-compose.yml

Run

```
docker compose up
```

Stop

```
docker compose down
```

Docker main commands

Category	Command	Description
Information	<code>docker --version</code>	Show Docker version
	<code>docker info</code>	Show system-wide information
	<code>docker help</code>	Show available commands

Docker main commands

Category	Command	Description
Images	<code>docker pull <image></code>	Download an image from Docker Hub
	<code>docker images</code>	List downloaded images
	<code>docker rmi <image></code>	Remove an image
	<code>docker build -t <name> .</code>	Build an image from a Dockerfile

Docker main commands

Category	Command	Description
Containers (Run/Start)	<code>docker run <image></code>	Run a container
	<code>docker run -it <image> bash</code>	Run an interactive container
	<code>docker run -d <image></code>	Run a container in background (detached)
	<code>docker run -p host:container <image></code>	Map a port
	<code>docker run --name <name> <image></code>	Start container with a custom name

Docker main commands

Category	Command	Description
Containers (Lifecycle)	<code>docker ps</code>	List running containers
	<code>docker ps -a</code>	List all containers (including stopped)
	<code>docker stop <container></code>	Stop a running container
	<code>docker start <container></code>	Start an existing container
	<code>docker restart <container></code>	Restart a container
	<code>docker rm <container></code>	Remove a container

Docker main commands

Category	Command	Description
Docker Compose	docker compose up	Start services from docker-compose.yml
	docker compose down	Stop and remove services
	docker compose logs	Show logs for all services
	docker compose ps	List compose-managed containers